VTPEH 6105

Lecture 06

HYPOTHESIS TESTING: COMPARISONS OF PREVALENCES ACROSS GROUPS

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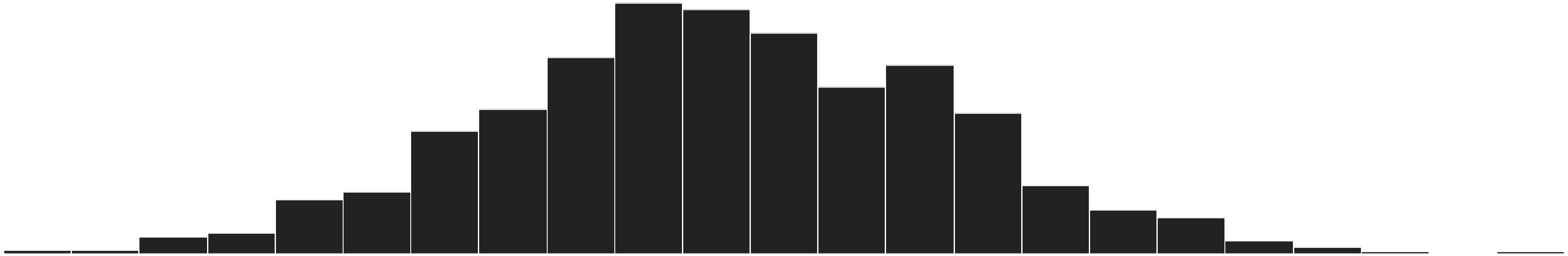


House Keeping

Feedback on Previous Homework

- Good overall (=
- Points lost on not answering all the questions, or not including the answer
- There will be no regrading

  One homework dropped at the end of the semester

  Help available at office hours

Comparison of Prevalences across Groups

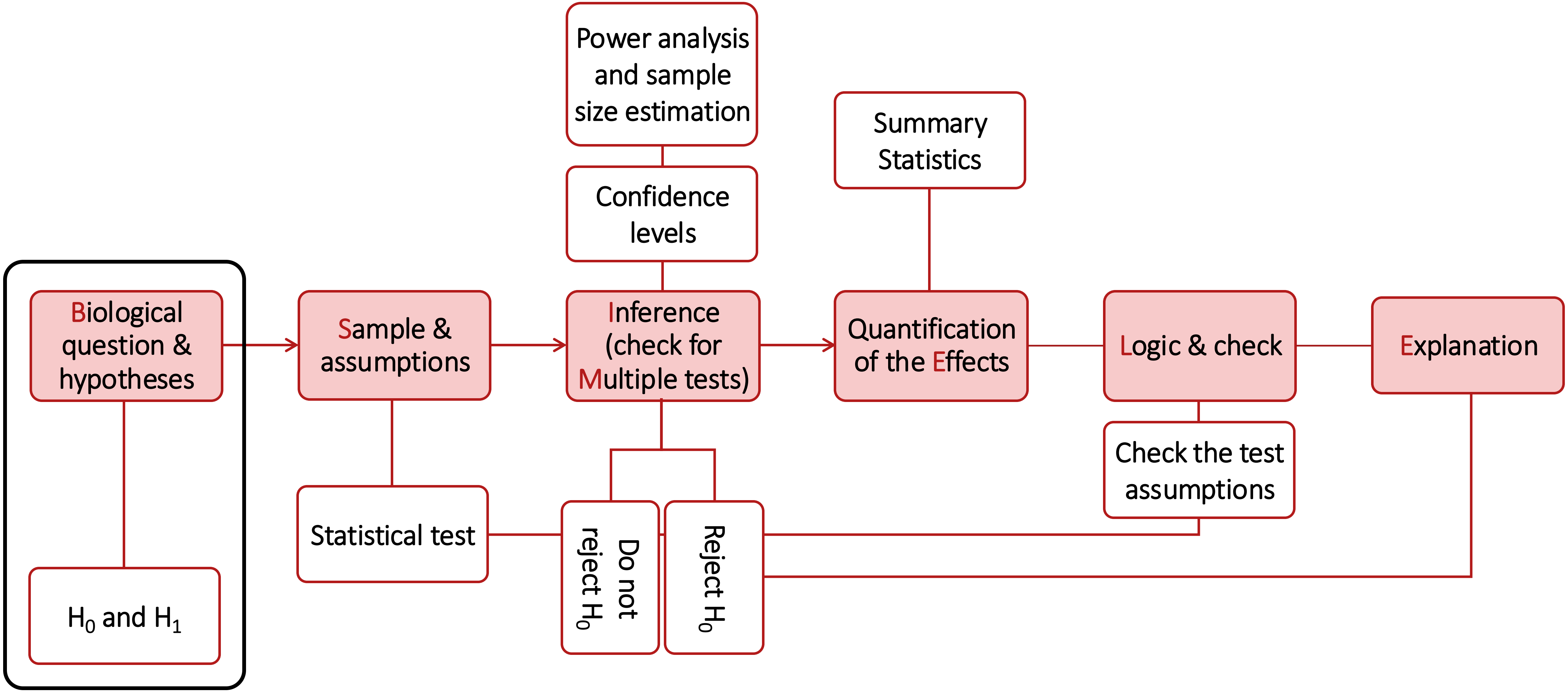
The basics with the χ^2 (chi-square) test

Considerations

B-SIMPLE

- **Biology and research question** What am I actually trying to answer? Does this make biological sense?
- **Sample and assumptions** Which approach can I reasonably use?
- **Inference power and significance** Can we, and do we, detect an association?
- **Multiple comparisons** Do the thresholds need to be adjusted to account for multiple testing?
- **Practical effect** What is the magnitude of the association?
- **Logic check** Does the output make sense?
- **Explanation** How do I communicate the results to my audience?

Hypothesis Testing Workflow



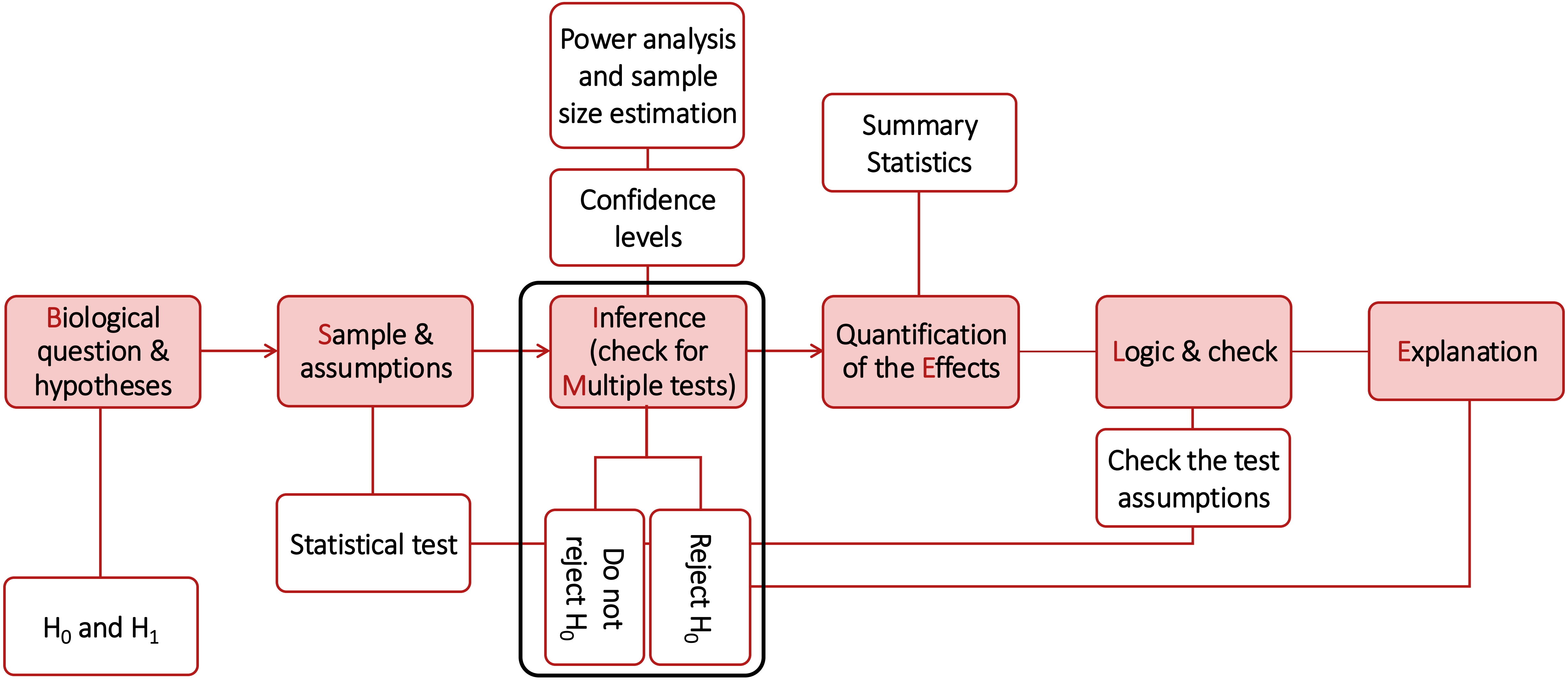
Working hypotheses

- Applied to explore associations between 2 categorical variables

- Null hypothesis (H_0): $\mu_1 = \mu_2$
  - Cancer prevalence is equal across smokers and non-smokers
  - There is no association between cancer and smoking

- Alternative hypothesis (H_1): $\mu_1 \neq \mu_2$
  - Cancer prevalence is different across smokers and non-smokers
  - There is an association between cancer and smoking

Hypothesis Testing Workflow



Chi2 – How?

- It all starts with a contingency table
  Efficient way to summarize prevalence data

	Cancer	No cancer
Nonsmoking	16	130
Smoking	105	49

Chi2 – How?

- It all starts with a contingency table
  Efficient way to summarize prevalence data

- Adding margins help calculating prevalences…
  And the distribution of individuals across groups expected under the null hypothesis

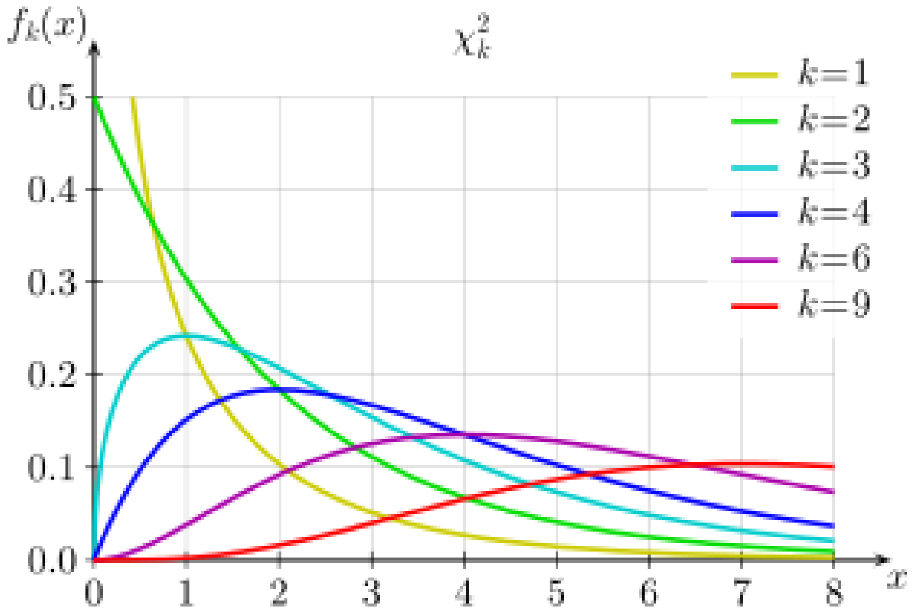
	Cancer	**No cancer**	*Total*
Nonsmoking	16	130	*146*
Smoking	105	49	*154*
Total	*121*	*179*	*300*

Chi2 – How?

- $\chi^2 = \sum \frac{(O-E)^2}{E}$

  With:
  - O the observed sample sizes within each group
  - E the expected sample sizes within each group under the null hypothesis
  - Summed across all groups (i.e., all combinations of the variables X and Y

I.e., it compares the observed distribution of individuals across groups, to the one expected under the null hypothesis



- High χ^2 values have low probabilities of being observed under the null hypothesis (p-values)

Chi2 – How?

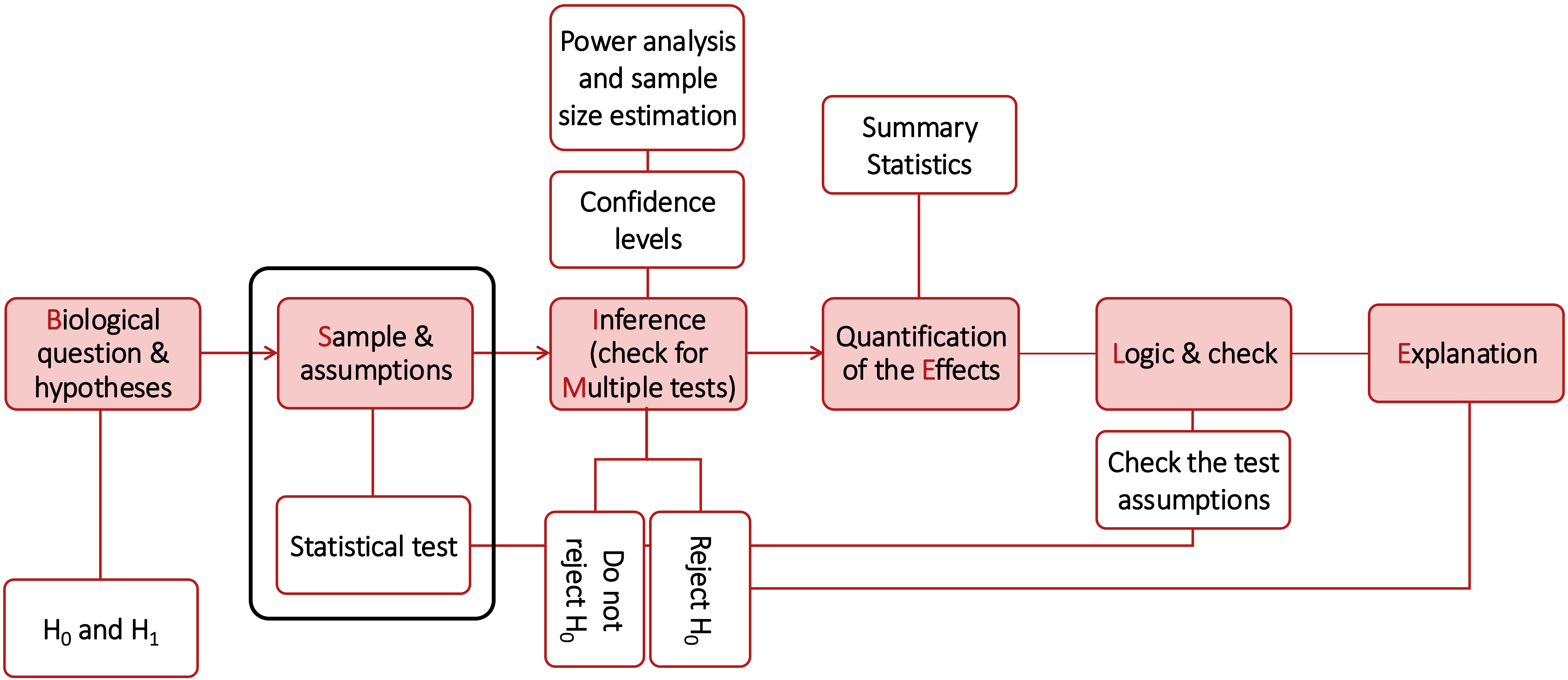
- $\chi^2 = \sum_{i,j} \frac{(O_{i,j} - E_{i,j})^2}{E_{i,j}}$

  With:
  - O the observed sample sizes within each group
  - E the expected sample sizes within each group under the null hypothesis
  - Summed across all groups (i.e., all combinations of the variables X and Y

Observed	**Cancer**	**No cancer**	*Total*
Nonsmoking	16	130	*146*
Smoking	105	49	*154*
Total	*121*	*179*	*300*

Expected under H_0	**Cancer**	**No cancer**	*Total*
Nonsmoking	$146 \times 121 / 300$ = 59	$146 \times 179 / 300$ = 87	*146*
Smoking	$154 \times 121 / 300$ = 62	$154 \times 179 / 300$ = 9q	*154*
Total	*121*	*179*	*300*

Hypothesis Testing Workflow



Chi2 – When? Assumptions

- Each group's samples must be **independent** from the other group

Study design

- The expected sample sizes in each group under the null hypothesis should be large enough (e.g., > 5)

Part of the chi2 calculation

Practice Time

Post-Hoc Tests, Multiple Comparisons, and P-Value Adjustment

Chi2 on more than 2 groups

- Work for more than 2 groups
  - If significant, should be followed by a post-hoc test
    E.g., pairwise chi2 with a Bonferroni correction

Other Approaches

Approaches Not Covered Here

- Fischer exact text
  - Appropriate when the expected sample size in each group under the null hypothesis is small (e.g., < 5)
  - Based on the calculation of the exact p-value based on the probability to observe each contingency table under the null hypothesis (based on the binomial distribution)

Before Next Time (Friday March 6, in LH4)

- Practice: Homework #06 due by **Thursday 5 PM**
- Use the Guiding Quiz #06 if needed
- Check you are ready for next class: Knowledge Check #06 due by **Friday 11 AM**

Need Help?

- Come to VTPEH 6108: Public Health Data Lab
  - Post your questions on the Discussion
- Come to office hours
  - **Tuesdays, 09:00-10:00 AM** (Jordan, Maddy)
  - **Wednesdays, 05:00-06:00 PM** (Mainul, Princal, Daniel and Harsh)
  - If you cannot make it at these times, please contact the course assistant of your choice by email
  - In S1-122, unless indicated otherwise on Canvas